# 生物统计和生物信息引论

## Lecture 7

## 生存分析：基本概念

王健桥
jianqw@mail.tsinghua.edu.cn

# Example: NCCTG Lung Cancer Data



- survival::lung in R
- The North Central Cancer Treatment Group (NCCTG) in the United States.
- **228 patients** with advanced lung cancer.
- Target outcome: Time (survival time, in days)
- Covariates: age, sex, etc

| | inst | time | status | age | sex | ph.ecog | ph.karno | pat.karno | meal.cal | wt.loss |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 306 | 2 | 74 | 1 | 1 | 90 | 100 | 1175 | NA |
| 2 | 3 | 455 | 2 | 68 | 1 | 0 | 90 | 90 | 1225 | 15 |
| 3 | 3 | 1010 | 1 | 56 | 1 | 0 | 90 | 90 | NA | 15 |
| 4 | 5 | 210 | 2 | 57 | 1 | 1 | 90 | 60 | 1150 | 11 |
| 5 | 1 | 883 | 2 | 60 | 1 | 0 | 100 | 90 | NA | 0 |
| 6 | 12 | 1022 | 1 | 74 | 1 | 1 | 50 | 80 | 513 | 0 |
| 7 | 7 | 310 | 2 | 68 | 2 | 2 | 70 | 60 | 384 | 10 |
| 8 | 11 | 361 | 2 | 71 | 2 | 2 | 60 | 80 | 538 | 1 |
| 9 | 1 | 218 | 2 | 53 | 1 | 1 | 70 | 80 | 825 | 16 |
| 10 | 7 | 166 | 2 | 61 | 1 | 2 | 70 | 70 | 271 | 34 |
| 11 | 6 | 170 | 2 | 57 | 1 | 1 | 80 | 80 | 1025 | 27 |
| 12 | 16 | 654 | 2 | 68 | 2 | 2 | 70 | 70 | NA | 23 |
| 13 | 11 | 728 | 2 | 68 | 2 | 1 | 90 | 90 | NA | 5 |
| 14 | 21 | 71 | 2 | 60 | 1 | NA | 60 | 70 | 1225 | 32 |
| 15 | 12 | 567 | 2 | 57 | 1 | 1 | 80 | 70 | 2600 | 60 |
| 16 | 1 | 144 | 2 | 67 | 1 | 1 | 80 | 90 | NA | 15 |
| 17 | 22 | 613 | 2 | 70 | 1 | 1 | 90 | 100 | 1150 | -5 |

# Why Survival Analysis



- Typical settings: time-to-event outcomes (survival time)
  - For example, time to death/recurrence
- Question 1: Time-to-event estimation/Distribution modeling
  - What is the distribution of time until an event occurs?
- Question 2: Group comparison
  - Do different groups have different distributions?
- Question 3: Effect of covariates
  - How do covariates affect survival time?
- ……

# Why Survival Analysis



- Typical settings: time-to-event outcomes
  - For example, time to death/recurrence
- Why not linear regression?
- Why not logistic regression?
- Key challenges in time-to-event data: censoring
  - Censoring: the event time is not observed (删失 in Chinese)
- Although the exact survival time is not observed, we have partial information about it.

# Example: NCCTG Lung Cancer Data



- survival::lung in R
- Target outcome: Time (survival time, in days)
- Covariates: age, sex, etc
- Status: censoring status, 1=censored, 2=dead
- 63 of 228 patients are censored.
- We don't know their actual survival time. We only know it is "at least X days."

| | inst | time | status | age | sex | ph.ecog | ph.karno | pat.karno | meal.cal | wt.loss |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 306 | 2 | 74 | 1 | 1 | 90 | 100 | 1175 | NA |
| 2 | 3 | 455 | 2 | 68 | 1 | 0 | 90 | 90 | 1225 | 15 |
| 3 | 3 | 1010 | 1 | 56 | 1 | 0 | 90 | 90 | NA | 15 |
| 4 | 5 | 210 | 2 | 57 | 1 | 1 | 90 | 60 | 1150 | 11 |
| 5 | 1 | 883 | 2 | 60 | 1 | 0 | 100 | 90 | NA | 0 |
| 6 | 12 | 1022 | 1 | 74 | 1 | 1 | 50 | 80 | 513 | 0 |
| 7 | 7 | 310 | 2 | 68 | 2 | 2 | 70 | 60 | 384 | 10 |
| 8 | 11 | 361 | 2 | 71 | 2 | 2 | 60 | 80 | 538 | 1 |
| 9 | 1 | 218 | 2 | 53 | 1 | 1 | 70 | 80 | 825 | 16 |
| 10 | 7 | 166 | 2 | 61 | 1 | 2 | 70 | 70 | 271 | 34 |
| 11 | 6 | 170 | 2 | 57 | 1 | 1 | 80 | 80 | 1025 | 27 |
| 12 | 16 | 654 | 2 | 68 | 2 | 2 | 70 | 70 | NA | 23 |
| 13 | 11 | 728 | 2 | 68 | 2 | 1 | 90 | 90 | NA | 5 |
| 14 | 21 | 71 | 2 | 60 | 1 | NA | 60 | 70 | 1225 | 32 |
| 15 | 12 | 567 | 2 | 57 | 1 | 1 | 80 | 70 | 2600 | 60 |
| 16 | 1 | 144 | 2 | 67 | 1 | 1 | 80 | 90 | NA | 15 |
| 17 | 22 | 613 | 2 | 70 | 1 | 1 | 90 | 100 | 1150 | -5 |

# Problem with Traditional Models



- Logistic Regression
  - Model: Survival status ~ Covariates
  - Information Loss: It ignores the survival time.
  - It treats a patient who died after 5 days the same as one who died after 500 days—both are simply labeled as “dead”.

# Problem with Traditional Models



- Linear Regression
  - Model: Survival time ~ Covariates
  - Information Loss: It ignores the censoring status.
  - For the 63 censored patients, their true survival time is unknown (we only know it is > X).
  - Strategy 1: treat censored time as event time. It leads to underestimation of the true event time.
  - Strategy 2: discard these individuals directly. Dropping 27% of cases leaves mainly high-risk patients who died early, leading to selection bias.

# Key Definitions in Survival Analysis






- For each unit $i$, there are two latent variables:
  - $T_i$: True time to event (e.g., death, failure).
  - $C_i$: Time to censoring (e.g., study ends, patient drops out).
- Observed time $Y_i = \min(T_i, C_i)$
- Survival time $T_i$
  - Time from a defined starting point to the occurrence of an event
  - Examples: time to death, time to AIDS
- Censoring time $C_i$
  - Time from a defined starting point to the end of observation without the event
  - Examples: study end, loss to follow-up, withdrawal

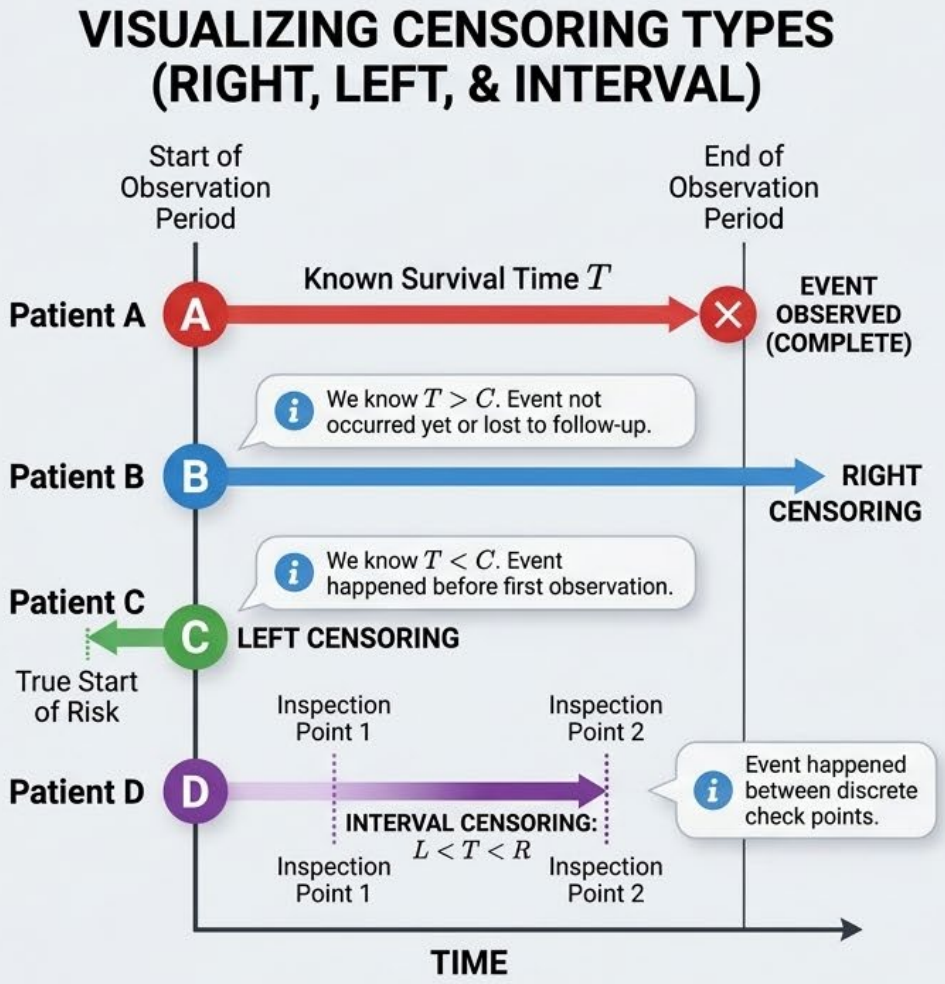
# Key Definitions in Survival Analysis



- Censoring indicator $\delta$
  - $\delta = 1$: event observed
  - $\delta = 0$: censored (event not observed)
- Imagine we are tracking how long a light bulb lasts.
  - Case A: The bulb burns out at 100 hours. Exact truth.
  - Case B: At 100 hours, the experiment ends, but the bulb is still glowing. Partial truth.

# Classifications of Censoring






## SUMMARY OF DATA TYPES

| Data Type | Definition | Mathematical Info | Real-World Example |
|---|---|---|---|
| Observed (Complete) | Event occurs within observation | $T$ is known | Clinical trial death |
| B Right Censoring | Event occurs after observation | $T > C$ | Patient alive at study end / lost to follow-up |
| C Left Censoring | Event occurs before observation | $T < C$ | Cancer detected upon first diagnosis |
| D Interval Censoring | Event occurs between two known times | $L < T < R$ | Bi-weekly health screenings |

The picture is generated by Nano Banana (Gemini)

# Classifications of Right-Censoring



- Independent Censoring:
  - Time-Fixed Censoring
    - Censored at a pre-specified time $C$.
    - "The experiment ends exactly at 5 PM on Friday."
  - Random Censoring
    - Censoring time is a random variable. $C_i$ is independent of $T_i$.
    - "Patients might move to another city or quit the study for reasons unrelated to their health."

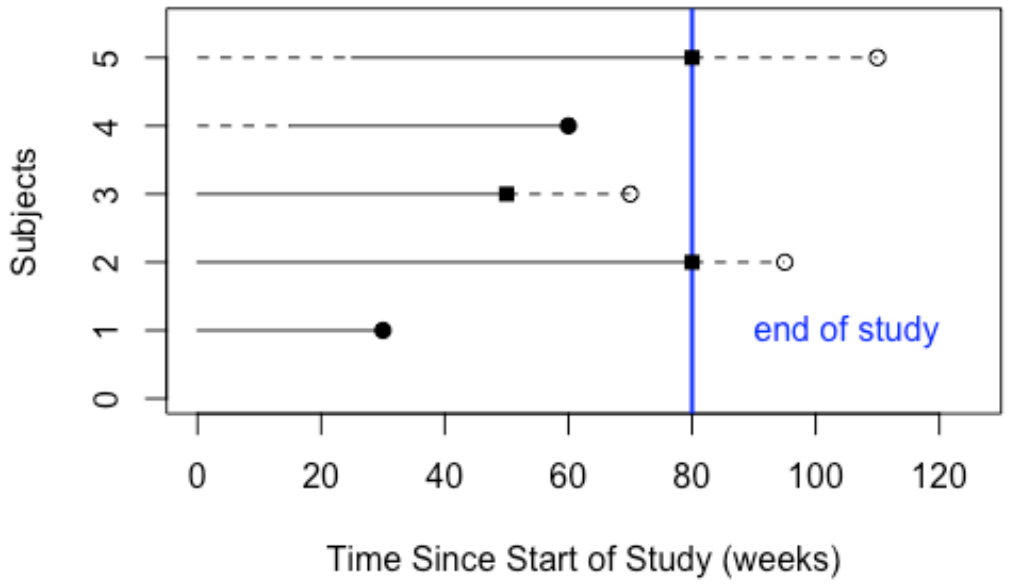
# Classifications of Right-Censoring



- Dependent Censoring:
  - Event-Fixed Censoring
    - Censored after a pre-specified number of events $r$.
    - "We stop the experiment as soon as the 3rd person dies."
    - $C = T_{(r)}$(The r-th ordered failure time). $C$ is directly determined by survival times $T$ of the subjects. NOT independent.

# An Example

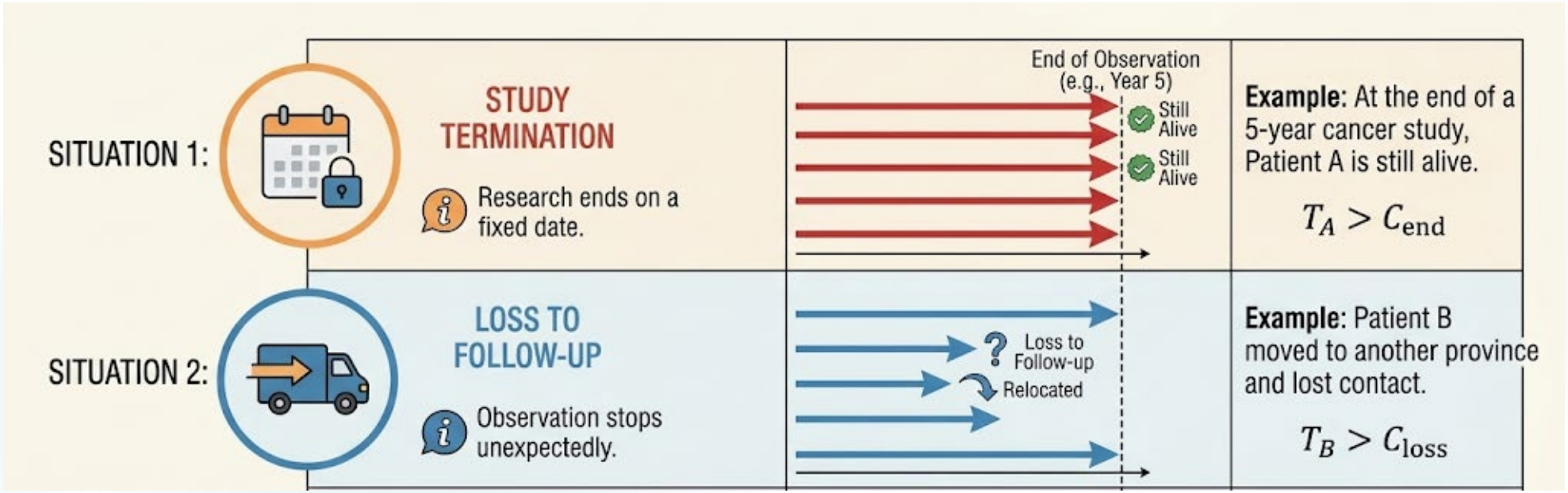





- 5. late entry and fixed right censoring
- 4. late entry
- 3. random-right censoring
- 2. fixed right censoring
- 1. uncensored

# Common Sources of Censoring






Independent Censoring

Independent Censoring

The picture is generated by Nano Banana (Gemini)

# Distribution of Time-to-Event



Recall that $T$ is a nonnegative variable denoting the time to the event of interest. The disbribution of $T$ could be discrete, continuous or a mixture of both. We focus on the **continuous distribution** here.

- Cumulative distribution function (CDF): $F(t) = \Pr(T \leq t)$.
- Probability density function (PDF): $f(t) = F'(t)$.
- Survival function (the probability that a subject survives **longer than** time $t$): $S(t) = \Pr(T > t) = 1 - F(t)$.

# Survival Function



The properties of $S(t)$:

- $S(0) = 1$.
- $S(\infty) = 0$.
- Non-increasing.
- If $T$ is a continuous random variable,

$$S(t) = \int_t^\infty f(u)\mathrm{d}u, \quad \text{and} \quad f(t) = -\frac{\mathrm{d}S(t)}{\mathrm{d}t}.$$

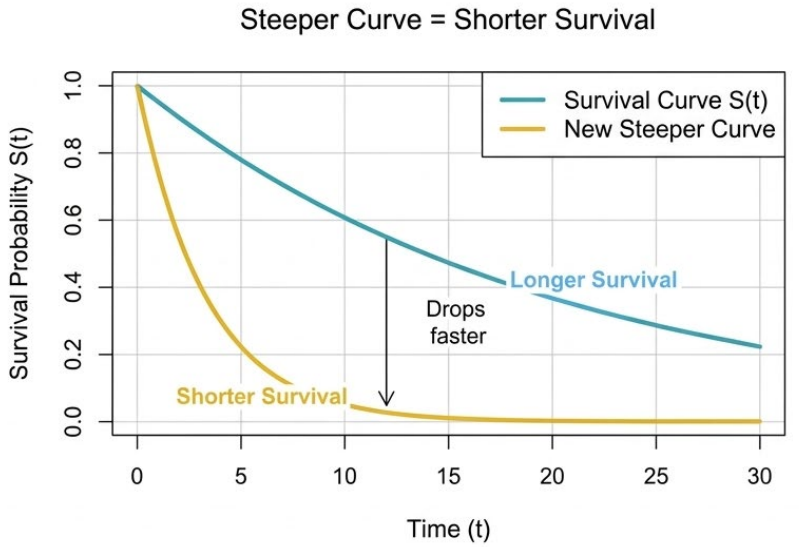
  There is a one-to-one correspondence between $f(t)$ and $S(t)$.

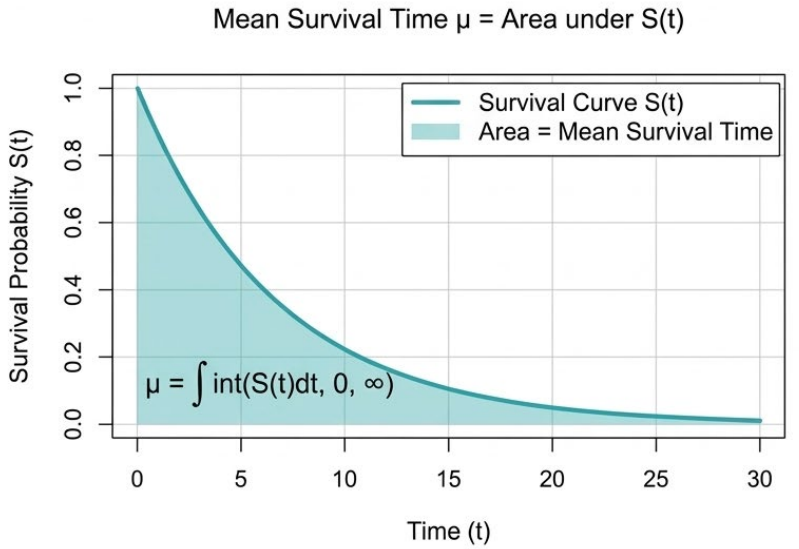
# Survival Function



- **Steeper curve correspond to shorter survival**

- **Mean survival time $\mu = E(T) =$ area under $S(t)$**



# Hazard Function



S(t) is not enough to describe 风险如何随时间变化

- 人类死亡率随年龄变化
- 疾病风险随时间变化（医学）

Hazard rate is the instantaneous rate of failure at time $t$ given that an individual is alive at time $t$.

$$\lambda(t) = \lim_{h \to 0} \frac{\Pr(t \leq T < t + h \mid T \geq t)}{h}.$$

Relationship:

$$\lambda(t) = \frac{\lim_{h \to 0} \frac{\Pr(t \leq T < t+h)}{h}}{\Pr(T \geq t)} = \frac{f(t)}{S(t)} = -\frac{S'(t)}{S(t)} = -\frac{\mathrm{d} \log\{S(t)\}}{\mathrm{d}t}.$$

# Cumulative Hazard Function



Cumulative hazard function is the total accumulated risk of an event occurring up to time $t$:

$$\Lambda(t) = \int_0^t \lambda(u)\mathrm{d}u, \quad \lambda(t) = -\frac{\mathrm{d}\log\{S(t)\}}{\mathrm{d}t} \implies \Lambda(t) = -\log\{S(t)\}.$$

Since $S(0) = 1$,

$$S(t) = \exp\{-\Lambda(t)\} = \exp\left\{-\int_0^t \lambda(u)\mathrm{d}u\right\}.$$

# Relationship Summary



$$f(t) = \frac{\mathrm{d}F(t)}{\mathrm{d}t}, \quad F(t) = \int_0^t f(s)\mathrm{d}s,$$

$$S(t) = 1 - F(t), \quad S(t) = \int_t^\infty f(s)\mathrm{d}s,$$

$$\lambda(t) = \frac{f(t)}{S(t)}, \quad \lambda(t) = \frac{\mathrm{d}\Lambda(t)}{\mathrm{d}t},$$

$$\Lambda(t) = \int_0^t \lambda(s)\mathrm{d}s,$$

Key relationship: $S(t) = \exp\{-\Lambda(t)\} = \exp\left\{-\int_0^t \lambda(u)\mathrm{d}u\right\}.$

# Q1: Distribution Modeling



- To estimate $S(t)$ (or equivalently, $F(t)$), we have two types of methods:
  - Non-parametric methods (discussed in this lecture)
    - Kaplan-Meier
    - Nelson-Aalen
  - Parametric methods (discussed in next lecture)

# Kaplan-Meier Estimator for survival function



| **Distinct Event Times** | $t_1$ | ... | $t_i$ | ... | $t_k$ |
|---|---|---|---|---|---|
| **# of Events** | $d_1$ | ... | $d_i$ | ... | $d_k$ |
| **# at Risk at $t_i$** | $n_1$ | ... | $n_i$ | ... | $n_k$ |

Decomposition of the survival function:

$$S(t) = \Pr(\text{survive at } t_1, ...t_i \leq t) = \prod_{i:t_i \leq t} \Pr(T > t_i \mid T \geq t_i).$$

$\Pr(T = t_i \mid T \geq t_i) \approx d_i/n_i$, which gives $\Pr(T > t_i \mid T \geq t_i) \approx 1 - d_i/n_i$.

# Kaplan-Meier Estimator



| Distinct Event Times | $t_1$ | … | $t_i$ | … | $t_k$ |
|---|---|---|---|---|---|
| # of Events | $d_1$ | … | $d_i$ | … | $d_k$ |
| # at Risk at $t_i$ | $n_1$ | … | $n_i$ | … | $n_k$ |

- **Kaplan-Meier estimator $\hat{S}$(t) (Kaplan & Meier, 1958, JASA):**

$$\hat{S}(t) = \prod_{i:t_i \le t} \frac{n_i - d_i}{n_i}$$

- $t_i$ :The distinct event times up to time $t$
- $n_i$ : The number of individuals at risk just prior to time $t_i$.
- $d_i$: The number of events (e.g., deaths) at time $t_i$.

# Example for Kaplan-Meier Estimator



- **When censoring exists**
  - Remission times (weeks) for 21 leukemia(白血病) patients in the treatment group are (+ indicates right censored):

| ID | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RT | 6 | 6 | 6 | 6+ | 7 | 9+ | 10 | 10+ | 11+ | 13 | 16 | 17+ | 19+ | 20+ | 22 | 23 | 25+ | 32+ | 32+ | 34+ | 35+ |

**Please fill the KM table**

**Notice: only consider event times (6, 7, 10, 13, 16, 22, 23)**

# Example for Kaplan-Meier Estimator



| $t_i$ | $n_i$ | $d_i$ | | $\hat{S}(t)$ |
|---|---|---|---|---|
| t=6 | 21 | 3 | 1.000×(21−3)/21 | 0.857 |
| t=7 | 17 | 1 | 0.857×(17−1)/17 | 0.807 |
| t=10 | 15 | 1 | 0.807×(15−1)/15 | 0.753 |
| t=13 | 12 | 1 | 0.753×(12−1)/12 | 0.690 |
| t=16 | 11 | 1 | 0.690×(11−1)/11 | 0.627 |
| t=22 | 7 | 1 | 0.627×(7−1)/7 | 0.538 |
| t=23 | 6 | 1 | 0.538×(6−1)/6 | 0.448 |

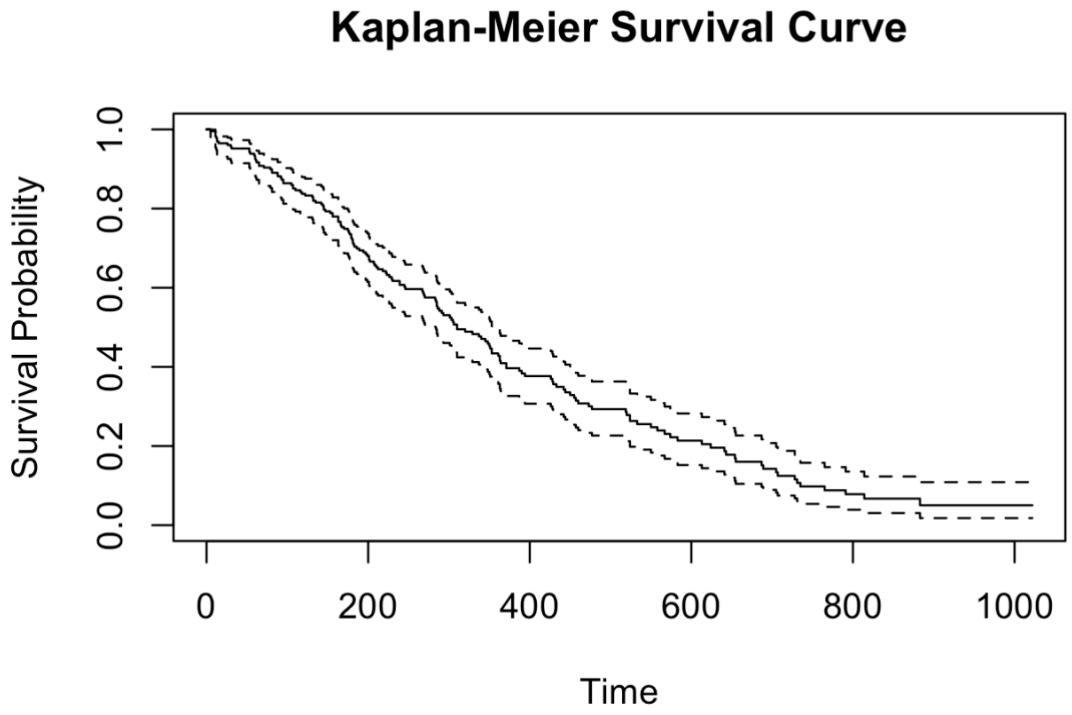
$$\hat{S}(t) = \prod_{i:t_i \le t} \frac{n_i - d_i}{n_i}$$

# Survival Analysis in R



- Data: NCCTG Lung Cancer Data

```r
# install.packages("survival")
library(survival)
lung
fit_km <- survfit(Surv(time, status) ~ 1, data = lung, conf.type = "log-log")
summary(fit_km)
plot(fit_km, xlab = "Time", ylab = "Survival Probability", main = "Kaplan-Meier Survival Curve")
```



# Variance Estimation of KM Estimator



Notice that

$$\log \hat{S}(t) = \sum_{i:t_i \le t} \log\left(\frac{n_i - d_i}{n_i}\right).$$

Assuming events at each time point $t$ are independent, the variance of the sum is

$$\mathrm{var}\{\log \hat{S}(t)\} = \sum_{i:t_i \le t} \mathrm{var}\left\{\log\left(\frac{n_i - d_i}{n_i}\right)\right\}.$$

Delta method gives

$$\mathrm{var}\left\{\log\left(\frac{n_i - d_i}{n_i}\right)\right\} = \left\{\frac{1}{(n_i - d_i)/n_i}\right\}^2 \mathrm{var}\left(\frac{n_i - d_i}{n_i}\right).$$

# Variance Estimation of KM Estimator



Since $d_i$ is approximately binomial with parameters $(n_i, p_i)$, where $p_i$ is the conditional failure probability at time $t_i$, we have

$$\text{var}\left(\frac{n_i - d_i}{n_i}\right) \approx \frac{d_i(n_i - d_i)}{n_i^3}.$$

Therefore, $\text{var}\{\log \hat{S}(t)\} = \sum_{i:t_i \le t} d_i/\{n_i(n_i - d_i)\}$. Finally, applying the delta method to the transformation $\hat{S}(t) = \exp\{\log \hat{S}(t)\}$, we obtain

$$\widehat{\text{var}}\{\hat{S}(t)\} \approx \hat{S}^2(t)\,\widehat{\text{var}}\{\log \hat{S}(t)\} = \hat{S}^2(t) \sum_{i:t_i \le t} \frac{d_i}{n_i(n_i - d_i)},$$

which is Greenwood's formula.

# Pointwise Confidence Interval



**Plain approach.** A direct normal-approximation confidence interval is then

$$\hat{S}(t) \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{var}}\{\hat{S}(t)\}}.$$

Although simple, this interval may perform poorly in finite samples and can even produce limits outside the admissible range $[0, 1]$.

```
survival::survfit(..., conf.type = "plain")
```

# Pointwise Confidence Interval



**Log transformation.** CI for $\Lambda(t) = -\log\{S(t)\}$ is given by (delta method)

$$-\log\{\hat{S}(t)\} \pm z_{\alpha/2}\sqrt{\frac{\widehat{\text{var}}\{\hat{S}(t)\}}{\{\hat{S}(t)\}^2}}.$$

CI for $S(t)$ is given by

$$\hat{S}(t) \times \left[\exp\left\{-z_{\alpha/2}\sqrt{\frac{\widehat{\text{var}}\{\hat{S}(t)\}}{\{\hat{S}(t)\}^2}}\right\}, \exp\left\{z_{\alpha/2}\sqrt{\frac{\widehat{\text{var}}\{\hat{S}(t)\}}{\{\hat{S}(t)\}^2}}\right\}\right].$$

Similar problem: $\Lambda(t) = -\log\{S(t)\} \in [0, \infty)$, but the CI can reach below 0.

```
survival::survfit(..., conf.type = "log")
```

# Pointwise Confidence Interval



**Log-log transformation.** A commonly preferred alternative is based on the log-log transformation $g\{S(t)\} = \log\{-\log S(t)\}$. Delta method gives

$$\widehat{\text{var}}[g\{S(t)\}] = \frac{1}{\{\log \hat{S}(t)\}^2} \sum_{i:t_i \le t} \frac{d_i}{(n_i - d_i)n_i}.$$

```
survival::survfit(..., conf.type = "log-log")
```

# Nelson-Aalen Estimator for Cumulative Hazard



| **Distinct Event Times** | $t_1$ | … | $t_i$ | … | $t_k$ |
|---|---|---|---|---|---|
| **# of Events** | $d_1$ | … | $d_i$ | … | $d_k$ |
| **# at Risk at $t_i$** | $n_1$ | … | $n_i$ | … | $n_k$ |

- NA estimator is used to estimate the cumulative hazard $\Lambda(t)$ and then estimate $S(t) = \exp\big(-\Lambda(t)\big)$
- Nelson-Aalen estimator for Cumulative Hazard $\hat{\Lambda}(t)$:

$$\hat{\Lambda}(t) = \sum_{i:t_i \le t} \frac{d_i}{n_i}$$

- Derived Survival Function: $\hat{S}_{NA}(t) = \exp\left(-\hat{\Lambda}(t)\right)$
- The variance estimator of NA estimator is: $\widehat{\mathrm{var}}\{\hat{\Lambda}(t)\} = \sum_{i:t_i \le t} \frac{d_i}{n_i^2}$

# Survival Analysis in R



- Data: NCCTG Lung Cancer Data

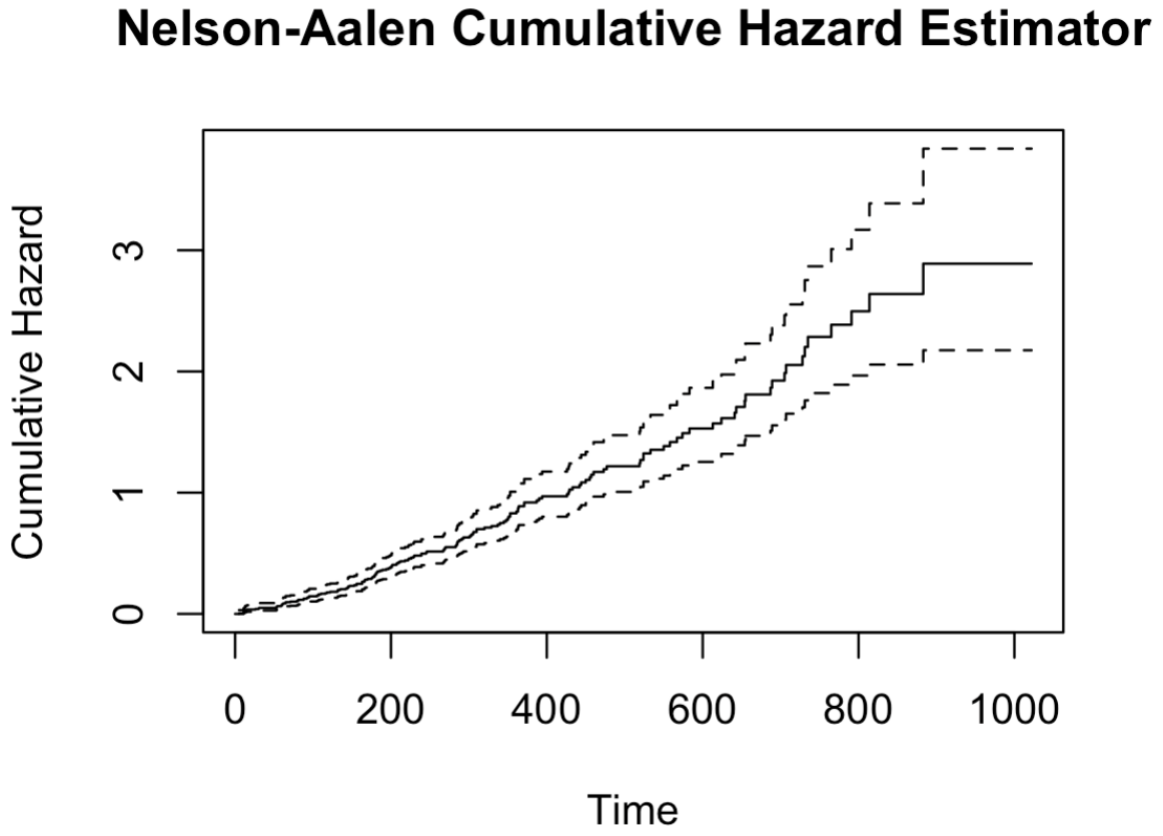
```
# N-A estimator
plot(fit_km, fun = "cumhaz", xlab = "Time", ylab = "Cumulative
Hazard", main = "Nelson-Aalen Cumulative Hazard Estimator")
```

**Usage**

```
## S3 method for class 'formula'
survfit(formula, data, weights, subset, na.action,
        stype=1, ctype=1, id, cluster, robust, istate, timefix=TRUE,
        etype, model=FALSE, error, entry=FALSE, time0=FALSE, ...)
```

**Arguments**

| | |
|---|---|
| formula | a formula object, which must have a Surv object as the response on the left of the ~ operator and, if desired, terms separated by + operators on the right. One of the terms may be a strata object. For a single survival curve the right hand side should be ~ 1. |
| data | a data frame in which to interpret the variables named in the formula, subset and weights arguments. |
| weights | The weights must be nonnegative and it is strongly recommended that they be strictly positive, since zero weights are ambiguous, compared to use of the subset argument. |
| subset | expression saying that only a subset of the rows of the data should be used in the fit. |
| na.action | a missing-data filter function, applied to the model frame, after any subset argument has been used. Default is options()$na.action. |
| stype | the method to be used estimation of the survival curve: 1 = direct, 2 = exp(cumulative hazard). |
| ctype | the method to be used for estimation of the cumulative hazard: 1 = Nelson-Aalen formula, 2 = Fleming-Harrington correction for tied events. |
| id | identifies individual subjects, when a given person can have multiple lines of data. |



# NA vs KM



Table 1: Comparison of Kaplan-Meier and Nelson-Aalen Estimators

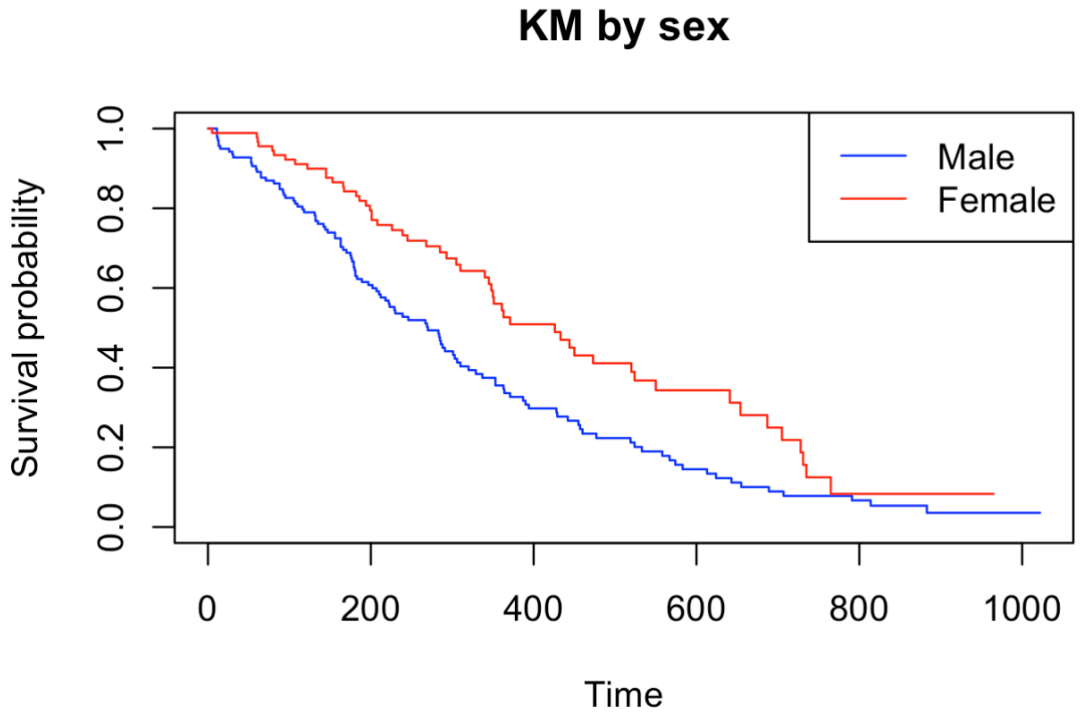
| **Feature** | **Kaplan–Meier Estimator** | **Nelson–Aalen Estimator** |
|---|---|---|
| **Primary Target** | Survival Function $S(t)$ | Cumulative Hazard Function $\Lambda(t)$ |
| **Core Formula** | $\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$ | $\hat{\Lambda}(t) = \sum_{t_i \le t} \frac{d_i}{n_i}$ |
| **Small Sample Behavior** | Can slightly underestimate survival probabilities in very small samples compared to NA. | Often performs better in small samples for hazard estimation. |

# Two Group Comparison



- Data: NCCTG Lung Cancer Data

```
12  fit_km_group <- survfit(Surv(time, status) ~ sex, data =
    lung, conf.type = "log-log")
13
14  plot(fit_km_group, col = c("blue", "red"), conf.int = FALSE
    , xlab = "Time", ylab = "Survival probability", main = "KM
    by sex")
15
16  legend("topright", legend = c("Male", "Female"), col = c
    ("blue", "red"), lty = 1)
```




- Is there any difference?

# Pooled Sample



- Non-parametric test to compare two populations
  - Collect two samples from each population
  - Construct a pooled sample with k distinct event times

| | Distinct Event Times | $t_1$ | ... | $t_i$ | ... | $t_k$ |
|---|---|---|---|---|---|---|
| Pool Sample | # of Events | $d_1$ | ... | $d_i$ | ... | $d_k$ |
| | # at Risk at $t_i$ | $n_1$ | ... | $n_i$ | ... | $n_k$ |
| Sample 1 | # of Events | $d_{11}$ | ... | $d_{1i}$ | ... | $d_{1k}$ |
| | # at Risk at $t_i$ | $n_{11}$ | ... | $n_{1i}$ | ... | $n_{1k}$ |
| Sample 2 | # of Events | $d_{21}$ | ... | $d_{2i}$ | ... | $d_{2k}$ |
| | # at Risk at $t_i$ | $n_{21}$ | ... | $n_{2i}$ | ... | $n_{2k}$ |

- Note $d_i = d_{1i} + d_{2i},\ n_i = n_{1i} + n_{2i},\ \ i = 1, \cdots, k$

# Log-Rank Test / Mantel-Haenszel Test



- $H_0: \lambda_1(t) = \lambda_2(t)$ $(S_1(t) = S_2(t))$
- $H_a: \lambda_1(t) \neq \lambda_2(t)$ for some t
- Consider $[t_i, t_i + \Delta)$ for small $\Delta$

| | Sample 1 | Sample 2 | Pooled Sample |
|---|---|---|---|
| # Events | $d_{1i}$ | $d_{2i}$ | $d_i$ |
| # Survivors at $t_i$ | $n_{1i} - d_{1i}$ | $n_{2i} - d_{2i}$ | $n_i - d_i$ |
| # Survivors right before $t_i$ | $n_{1i}$ | $n_{2i}$ | $n_i$ |
| # Expected events | $d_i n_{1i}/n_i$ | $d_i n_{2i}/n_i$ | $d_i$ |

- If $\lambda_1(t) \neq \lambda_2(t)$ around $t_i$, there should be association between sample and event in this $2 \times 2$ table (equivalently, $d_{1i}$ is far from $d_i n_{1i}/n_i$)

# Log-Rank Test: Procedure



Under $H_0$, $\mathbb{E}(d_{1i}) = d_i n_{1i}/n_i$.
Test statistic:

$$Z = \frac{\sum_{i=1}^{n}\{d_{1i} - \mathbb{E}(d_{1i})\}}{\sqrt{\sum_{i=1}^{n} \mathrm{var}(d_{1i})}} \approx \mathcal{N}(0,1), \quad \text{or equivalently,} \quad Z^2 \approx \chi_1^2.$$

Since $d_{1i} \mid (n_i, d_i, n_{1i}) \overset{H_0}{\sim} \text{Hypergeometric}(n_i, d_i, n_{1i})$,

$$P\left(d_{1i} = x \mid n_i, d_i, n_{1i}\right) = \frac{\binom{d_i}{x}\binom{n_i - d_i}{n_{1i} - x}}{\binom{n_i}{n_{1i}}},$$

$$\mathrm{var}(d_{1i}) = d_i \frac{n_{1i}}{n_i}\left(1 - \frac{n_{1i}}{n_i}\right)\frac{n_i - d_i}{n_i - 1}.$$

# Log Rank Test: A Simple Example



- The Scenario (n = 6)
- Group 1 (Treatment): 3 patients.
  - Survival times: 5 (Event), 8+ (Censored), 12 (Event)
- Group 2 (Control): 3 patients.
  - Survival times: 2 (Event), 3 (Event), 7 (Event)

- Test if the hazard functions of the two group are different?

# Log Rank Test: A Simple Example



| | Distinct Event Times | $t_1$ | ... | $t_i$ | ... | $t_k$ |
|---|---|---|---|---|---|---|
| Pool Sample | # of Events | $d_1$ | ... | $d_i$ | ... | $d_k$ |
| | # at Risk at $t_i$ | $n_1$ | ... | $n_i$ | ... | $n_k$ |
| Sample 1 | # of Events | $d_{11}$ | ... | $d_{1i}$ | ... | $d_{1k}$ |
| | # at Risk at $t_i$ | $n_{11}$ | ... | $n_{1i}$ | ... | $n_{1k}$ |
| Sample 2 | # of Events | $d_{21}$ | ... | $d_{2i}$ | ... | $d_{2k}$ |
| | # at Risk at $t_i$ | $n_{21}$ | ... | $n_{2i}$ | ... | $n_{2k}$ |

| | Distinct Event Times | $t_1$=2 | $t_2$=3 | $t_3$=5 | $t_4$=7 | $t_5$=12 |
|---|---|---|---|---|---|---|
| Pool Sample | # of Events ($d_i$) | 1 | 1 | 1 | 1 | 1 |
| | # at Risk at $t_i$ ($n_i$) | 6 | 5 | 4 | 3 | 1 |
| Sample 1 | # of Events ($d_{1i}$) | 0 | 0 | 1 | 0 | 1 |
| (Treatment) | # at Risk at $t_i$ ($n_{1i}$) | 3 | 3 | 3 | 2 | 1 |
| | # Expected ($E_{1i}$) | 0.5 | 0.6 | 0.75 | 0.667 | 1 |
| Sample 2 | # of Events ($d_{2i}$) | 1 | 1 | 0 | 1 | 0 |
| (Control) | # at Risk at $t_i$ ($n_{2i}$) | 3 | 2 | 1 | 1 | 0 |
| | # Expected ($E_{2i}$) | 0.5 | 0.4 | 0.25 | 0.333 | 0 |

$$E_{ji} = d_i \times \frac{n_{ji}}{n_i}, j = 1,2$$

# Log Rank Test: A Simple Example



$$
\begin{aligned}
U &= \sum_{i=1}^{k}\left[d_{1i} - \frac{d_i n_{1i}}{n_i}\right] \\
&= \left(0 - \frac{1\times 3}{6}\right) + \left(0 - \frac{1\times 3}{5}\right) + \left(1 - \frac{1\times 3}{4}\right) + \left(0 - \frac{1\times 2}{3}\right) + \left(1 - \frac{1\times 1}{1}\right) \\
&= (0 - 0.500) + (0 - 0.600) + (1 - 0.750) + (0 - 0.667) + (1 - 1.000) \\
&= -1.517
\end{aligned}
$$

$$
\begin{aligned}
\widehat{var}(U) &= \sum_{i=1}\left[\frac{d_i n_{1i}(n_i - d_i)(n_i - n_{1i})}{n_i^2(n_i - 1)}\right] \\
&= \left[\frac{1\cdot 3\cdot 5\cdot 3}{36\cdot 5}\right] + \left[\frac{1\cdot 3\cdot 4\cdot 2}{25\cdot 4}\right] + \left[\frac{1\cdot 3\cdot 3\cdot 1}{16\cdot 3}\right] + \left[\frac{1\cdot 2\cdot 2\cdot 1}{9\cdot 2}\right] + 0 \\
&= 0.2500 + 0.2400 + 0.1875 + 0.2222 + 0 \\
&= 0.8997
\end{aligned}
$$

1. Z statistic:

$$
Z = \frac{U}{\sqrt{\widehat{var}(U)}} = \frac{-1.517}{\sqrt{0.8997}} \approx -1.599
$$

Since Z = -1.599 falls within the standard normal distribution $N(0,1)$ non-rejection region (between -1.96 and 1.96 for $\alpha = 0.05$ ).

We fail to reject $H_0$ at level $\alpha = 0.05$.

# Log Rank Test: A Simple Example



2. Chi-square statistic:

$$\chi^2 = Z^2 = (-1.599)^2 \approx \mathbf{2.557}$$

Since $Z^2$ = 2.557 falls within the chi-square distribution with the degree of freedom non-rejection region (less than 3.84 for $\alpha = 0.05$ ).
We fail to reject $H_0$ at level $\alpha = 0.05$.

```
Call:
survdiff(formula = Surv(time, status) ~ group)

          N Observed Expected (O-E)^2/E (O-E)^2/V
group=A 3        2     3.52     0.654      2.56
group=B 3        3     1.48     1.551      2.56

 Chisq= 2.6  on 1 degrees of freedom, p= 0.1
```

# Example: NCCTG Lung Cancer Data



The data comes from the North Central Cancer Treatment Group (NCCTG) in the United States.

**Subjects:** It contains follow-up data on **228 patients** with advanced lung cancer.

If the female and males survival function are different?

| | inst | time | status | age | sex | ph.ecog | ph.karno | pat.karno | meal.cal | wt.loss |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 306 | 2 | 74 | 1 | 1 | 90 | 100 | 1175 | NA |
| 2 | 3 | 455 | 2 | 68 | 1 | 0 | 90 | 90 | 1225 | 15 |
| 3 | 3 | 1010 | 1 | 56 | 1 | 0 | 90 | 90 | NA | 15 |
| 4 | 5 | 210 | 2 | 57 | 1 | 1 | 90 | 60 | 1150 | 11 |
| 5 | 1 | 883 | 2 | 60 | 1 | 0 | 100 | 90 | NA | 0 |
| 6 | 12 | 1022 | 1 | 74 | 1 | 1 | 50 | 80 | 513 | 0 |
| 7 | 7 | 310 | 2 | 68 | 2 | 2 | 70 | 60 | 384 | 10 |
| 8 | 11 | 361 | 2 | 71 | 2 | 2 | 60 | 80 | 538 | 1 |
| 9 | 1 | 218 | 2 | 53 | 1 | 1 | 70 | 80 | 825 | 16 |
| 10 | 7 | 166 | 2 | 61 | 1 | 2 | 70 | 70 | 271 | 34 |
| 11 | 6 | 170 | 2 | 57 | 1 | 1 | 80 | 80 | 1025 | 27 |
| 12 | 16 | 654 | 2 | 68 | 2 | 2 | 70 | 70 | NA | 23 |

| | |
|---|---|
| inst: | Institution code |
| time: | Survival time in days |
| status: | censoring status 1=censored, 2=dead |
| age: | Age in years |
| sex: | Male=1 Female=2 |

# Example: NCCTG Lung Cancer Data

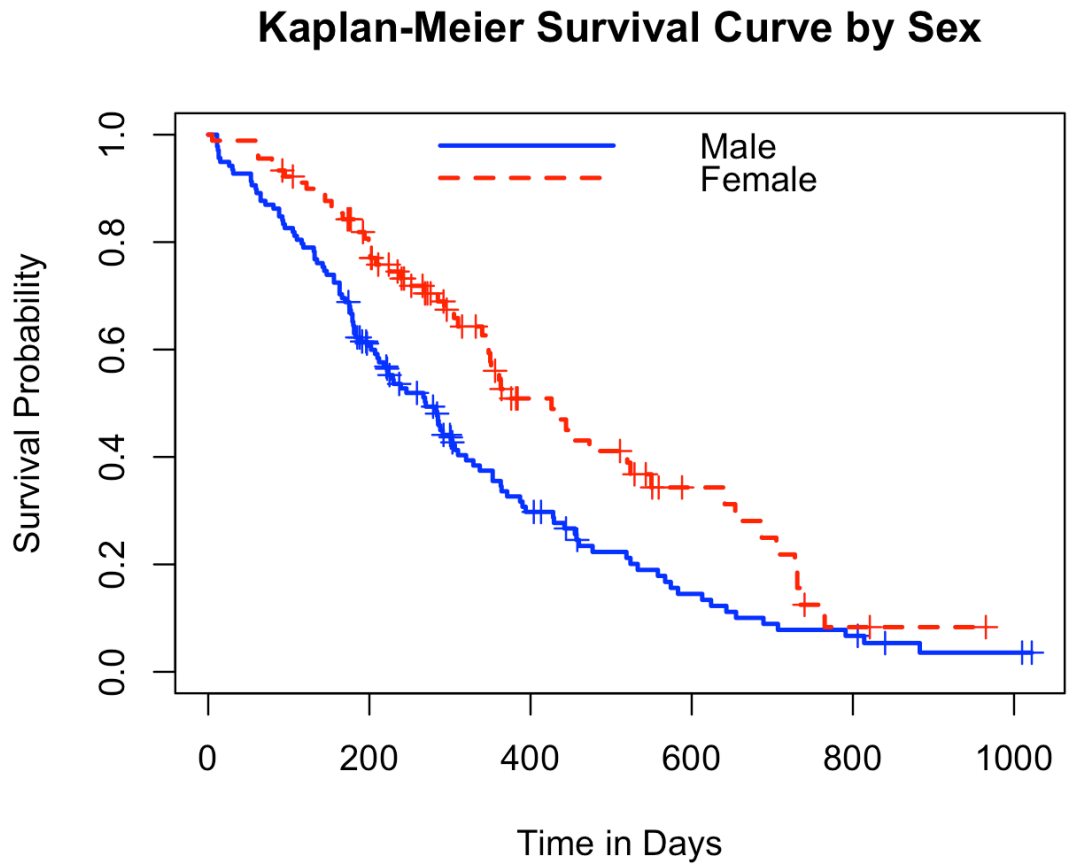


## KM estimator

```r
km_fit <- survfit(Surv(time, status) ~ sex, data = lung)

plot(km_fit,
     main = "Kaplan-Meier Survival Curve by Sex",
     xlab = "Time in Days",
     ylab = "Survival Probability",
     col = c("blue", "red"),
     lty = 1:2,
     lwd = 2,
     mark.time = TRUE,
     conf.int = FALSE
)

legend("topright",
       legend = c("Male", "Female"),
       col = c("blue", "red"),
       lty = 1:2,
       lwd = 2,
       bty = "n"
)
```



# Example: NCCTG Lung Cancer Data



Log-Rank test: if there is statistical difference between male and female.

```
log_rank_test <- survdiff(Surv(time, status) ~ sex, data = lung)

print(log_rank_test)
```

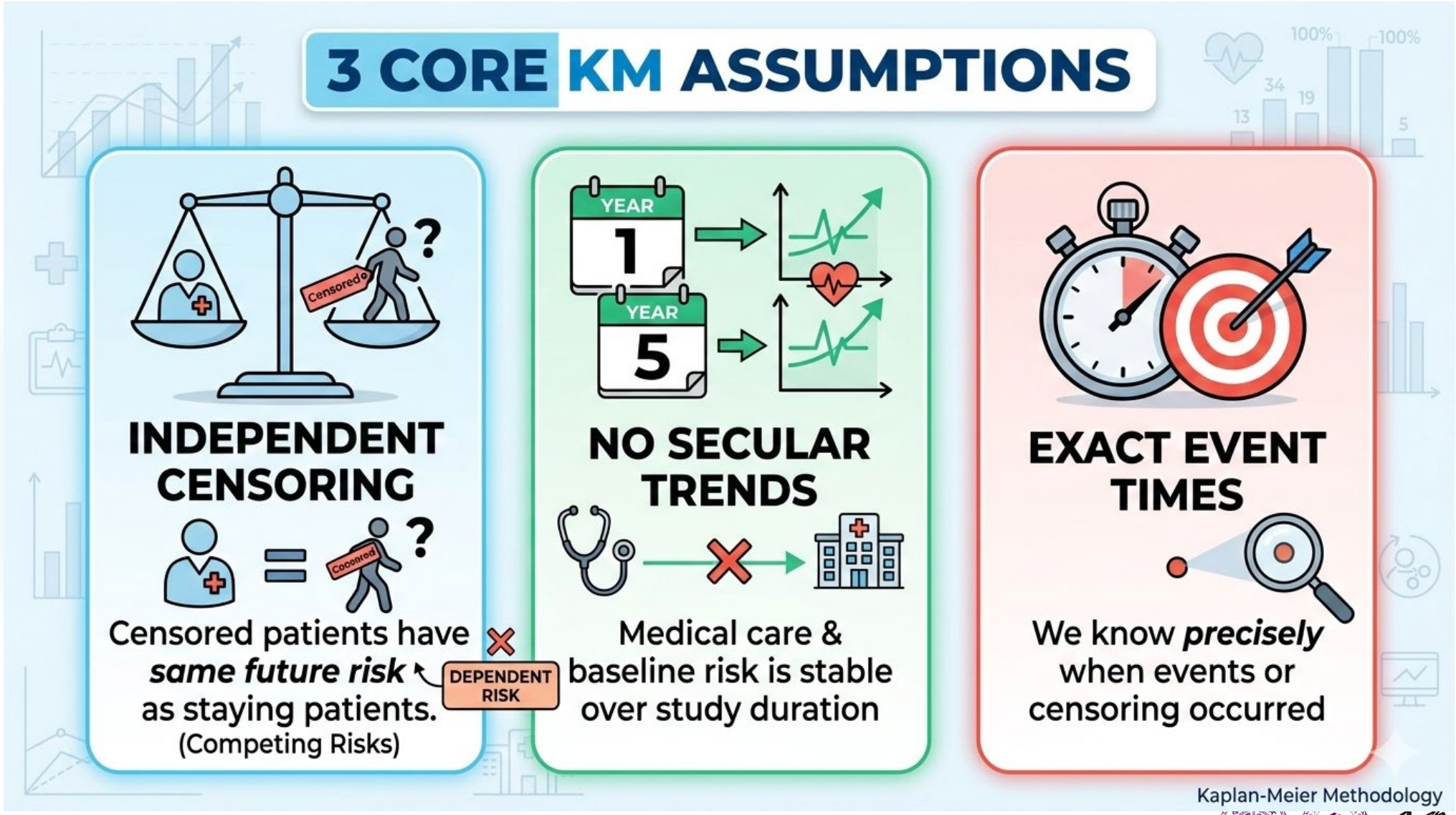
```
Call:
survdiff(formula = Surv(time, status) ~ sex, data = lung)

        N Observed Expected (O-E)^2/E (O-E)^2/V
sex=1 138      112     91.6      4.55      10.3
sex=2  90       53     73.4      5.68      10.3

 Chisq= 10.3  on 1 degrees of freedom, p= 0.001
```

# Assumptions for Kaplan-Meier Estimator

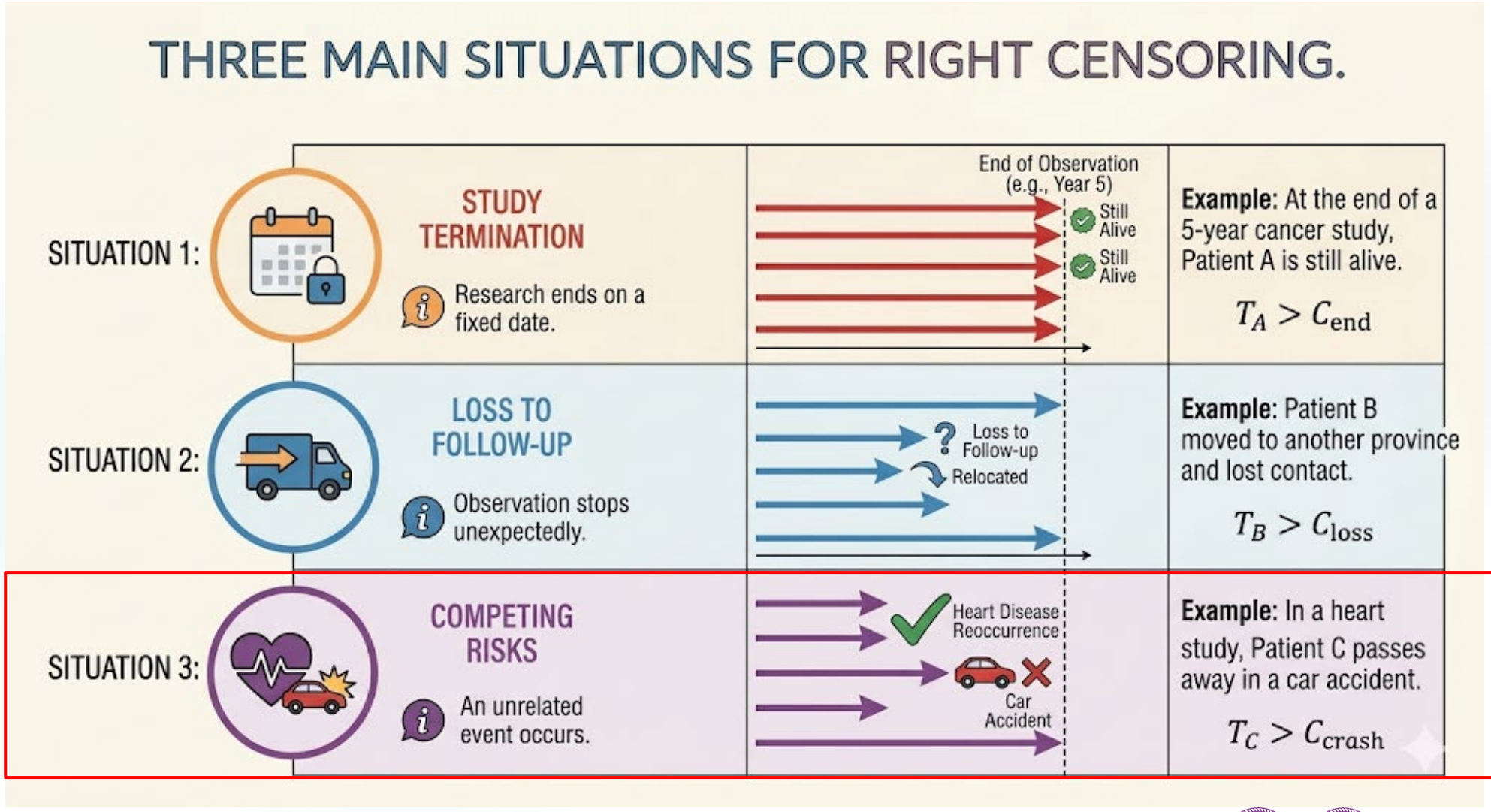





The picture is generated by Nano Banana (Gemini)

# Right-Censoring vs Competing Events



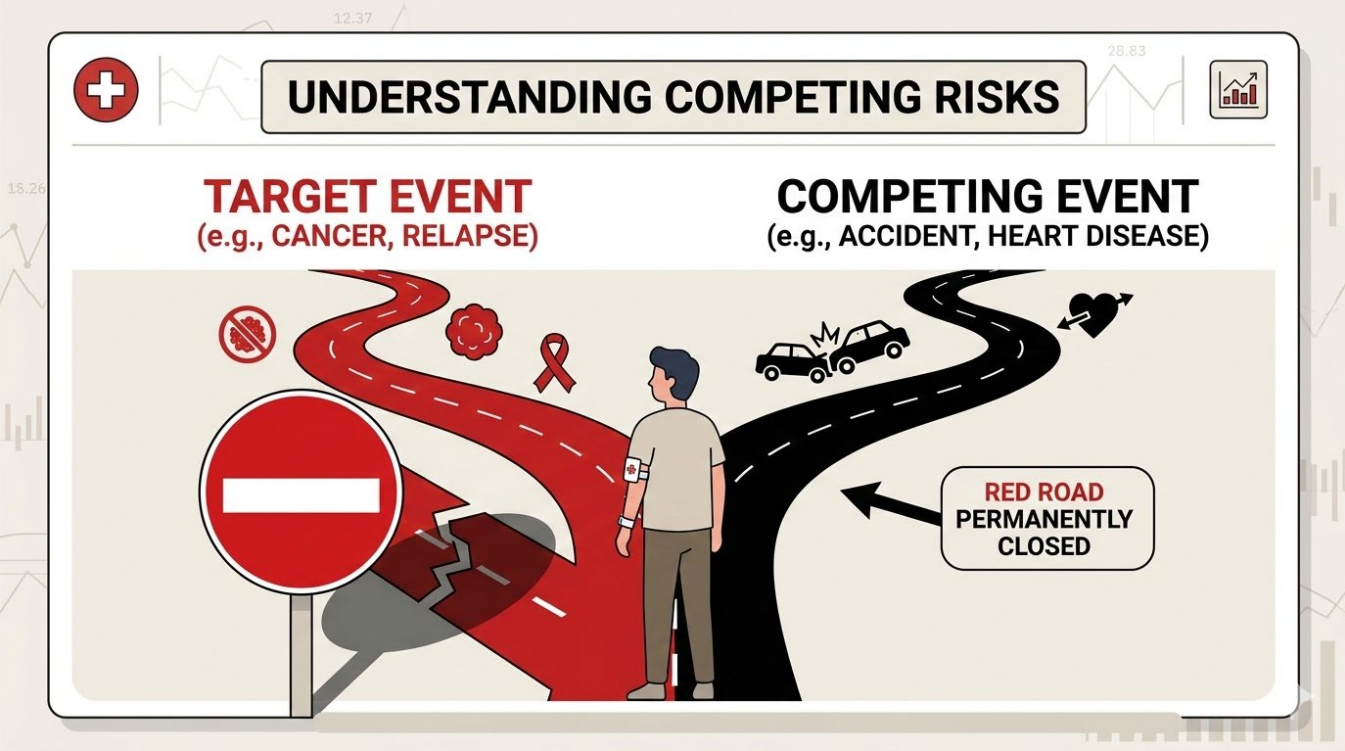



Independent Censoring

Independent Censoring

The picture is generated by Nano Banana (Gemini)

# Beyond Kaplan-Meier: The Competing Risks Dilemma




The picture is generated by Nano Banana (Gemini)

- The Reality: Patients has probability to die from cancer and other events (mutually exclusive).
- The Kaplan-Meier Flaw:
  - KM treats competing events (e.g., car crash) as "Censored".
- Fatal Assumption:
  - KM mathematically assumes these patients(die from accident) could still get cancer if we kept watching them!

overestimate the probability of the target event.

# Competing Risks Introduction



- Definition:
  - In standard survival analysis, we usually focus on a single event of interest. However, in the real world, a subject may experience multiple mutually exclusive events.
  - A competing risk is an event whose occurrence either prevents the event of interest from happening or fundamentally alters the probability of it occurring.

# The Problem with Kaplan-Meier



- Standard KM Assumption: Censoring is completely independent of the event of interest.
- The Competing Risk Problem:
  - Competing events ≠ Other censoring (e.g., lost to follow-up).
- A competing event permanently reduces the target event risk to zero.
- The Consequence:
  - Treating them equally violates the core assumption, causing KM to significantly overestimate the cumulative probability of the primary event.
- To properly estimate the probability of a specific event in the presence of competing risks, we use the **Cumulative Incidence Function (CIF)** instead of $1 - \hat{S}(t)$ (KM failure estimate).

# Solution: Cumulative Incidence Function



- **Definition:** The CIF for an event of type k estimates the marginal probability of experiencing event k by time t:
  - $CIF_k(t) = P(T \leq t, D = k) = \int_0^t \lambda_k(u) S_{overal}(u) du$
  - D = k: The specific cause of failure (e.g., k=1 for heart disease, k=2 for cancer).
  - $\lambda_k(u)$: The **cause-specific hazard function** for event k.
  - $S(u)$: The **overall survival function** (the probability of surviving *all* competing risks up to time u).
- **Property**
  - $\sum_k CIF_k(t) = 1 - S_{overal}(t)$

# The Uniqueness of Survival Analysis



- **Focuses on "Time-to-Event", Not Just "Yes/No"**
  - Standard classification models only ask *whether* an event occurred. Survival analysis analyzes the continuous time leading up to the event, answering both *if* and ***when***. Standard regression fails here because time data is strictly positive and highly skewed.
- **Elegantly Handles "Censored" Data**
  - In the real world, patients drop out or a study ends before everyone experiences the event. Instead of discarding these "incomplete" records, survival analysis mathematically incorporates censored subjects to maximize the value of all available data without introducing bias.
- **Evaluates Dynamic Risk via the "Hazard Function"**
  - Instead of static probabilities, it uses the Hazard Function $\lambda(t)$ to measure the *instantaneous* risk of an event happening right now, given that the subject has survived this long. It perfectly captures how real-world risks fluctuate over time.

# Summary



- Why Survival Analysis?
- Key Definitions: Survival Time, Censoring
- CDF, PDF, Survival Function, Hazard Function, Cumulative Hazard Function
- Kaplan-Meier Estimator
  - Assumptions, Method, Variance Estimator (Greenwood’s Formula), CI
- Nelson-Aalen Estimator
  - Difference with Kaplan-Meier Estimator
- Log-Rank Test
  - Method, Calculation
- Competing Risks Introduction